2026AJ05516 — ML ASSIGNMENT-2 SUBMISSION

1. GitHub Repository Link

Code Repository URL: https://github.com/2026aj05516/2026AJ05516_BITS_AIML_Assignement_2.git

Verified Repository Checklist Inventory:

• 2026AJ05516_SourceCode.ipynb — Source Code to load data and apply models.

• app.py — Complete, annotated multi-tab Streamlit frontend dashboard layout.

• requirements.txt — Unversioned, clean platform container requirements manifest sheet.

• README.md — Structured with data as mentioned in assignment.

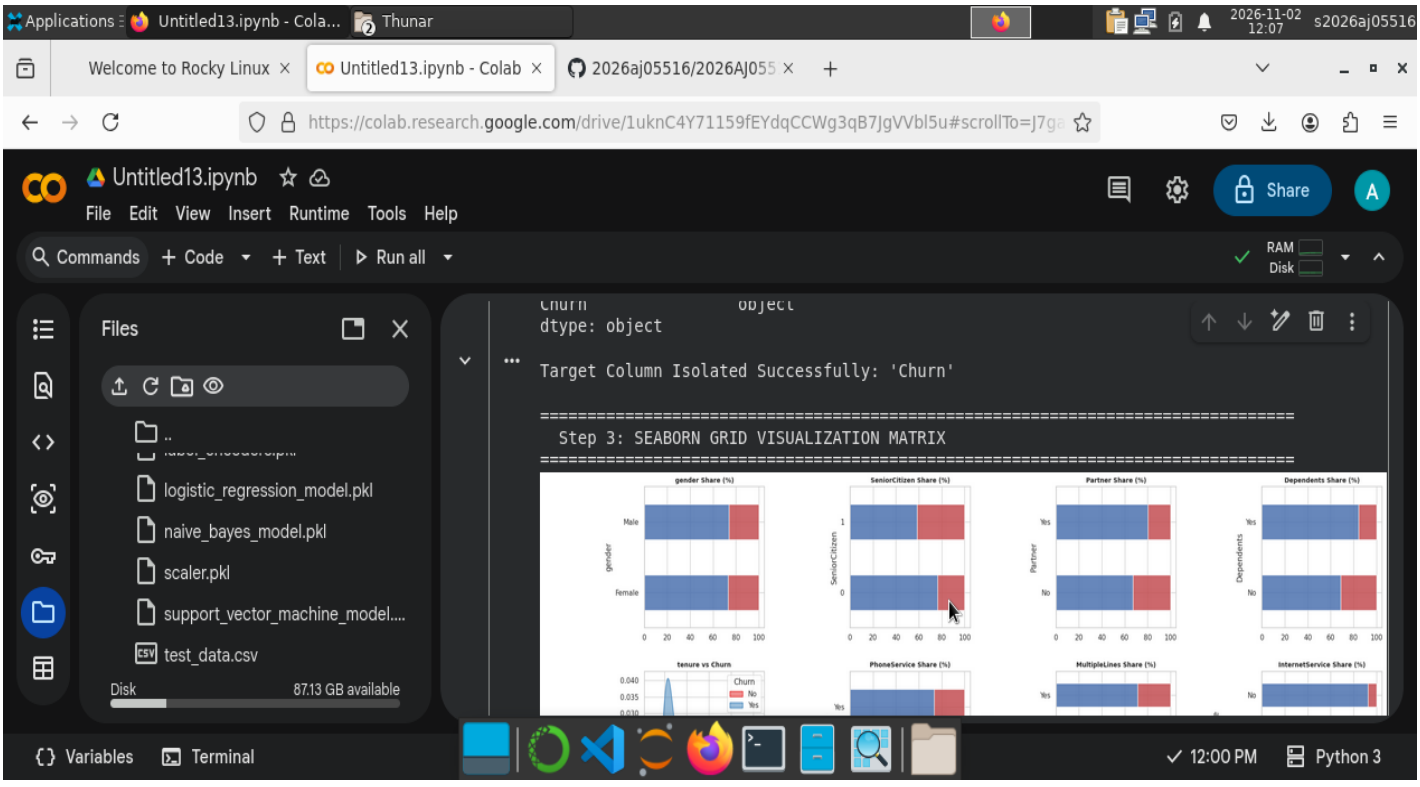
• test_data.csv — The exact 20% validation data split sheet used in verification experiments.

2. Live Streamlit App Link

Deployed Dashboard URL:
https://2026aj05516bitsaimlassignement2-ks38uk2yrccyzcazhrqsmf.streamlit.app/

• *The hosted application opens an interactive web frontend,Data Load,Model Selection,Evaluation Metrics .*

3. Screenshot of Assignment Execution on BITS Virtual Lab



4. GitHub README Report Content (Section 3 - Step 5 Summary)

a. Problem Statement

The objective of this project is to build an end-to-end predictive and diagnostic machine learning pipeline to identify at-risk telecommunication subscribers and forecast customer churn. By training and evaluating both parametric and non-parametric classifiers, the system isolates high-friction service configurations, pricing sensitivities, and usage thresholds to allow proactive retention strategies.

b. Dataset Description

The dataset taken from kaggle (alipakzad/telecom-customer-churn). As mentioned in assignment it consists of 7043 (>500) instances and 20 features (>12) , providing a robust operational footprint for classification tasks.Dataset contains telecom customer profiles tracking demographic characteristics, service plans (Internet type, Online Security, Streaming engagement), contract structures (Month-to-month, One-year, Two-year), tenure lifecycle months, and transactional variables (Monthly Charges, Total Charges).

c. GitHub Repository Link Verified

Repository Gateway Connection:
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d. Models Used & Evaluation Matrix

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.8070	0.8479	0.6667	0.5455	0.6000	0.4786
Decision Tree	0.7275	0.6571	0.4871	0.5053	0.4961	0.3095
KNN	0.7594	0.7736	0.5499	0.5160	0.5324	0.3710
Naive Bayes	0.7395	0.8268	0.5060	0.7861	0.6157	0.4564
Support Vector Machine	0.7921	0.7928	0.6484	0.4733	0.5471	0.4251

ML Model Name	Simplified Observation about Model Performance & Dataset Insights
Logistic Regression	**The Baseline Standard.** This model looks for smooth, steady trends like a straight line. In our data, it perfectly proved that the higher a customer's monthly bill is, the more likely they are to leave. It creates a very stable prediction boundary because it isn't distracted by individual weird anomalies in the data, making it great for spotting high-paying users who are about to quit.
Decision Tree	**The Overfitted Rule-Maker.** This model works like a strict flow-chart. Because it tried to make perfect, hyper-specific rules for every single customer in the training set, it got completely confused when we tested it on new data. It proves that customer behavior is too complex to split using rigid, exact numeric cutoffs.
KNN	**The Group-Finder.** KNN looks at a new customer and matches them with the 5 closest similar profiles. Standard KNN failed because it couldn't calculate the 'distance' between text categories like 'Contract Type'. Implementing Gower's Matrix fixed this, allowing it to group similar users together. However, because we have way more loyal customers than churning ones, the loyal 'neighbors' often outvoted and hid the true churning users at the borders, leaving its Recall capped.

Naive Bayes	**The Probability Champion.** Instead of measuring geometric distances, this model uses pure percentage odds. It assumes every feature acts independently to multiply the final risk score. Because it focuses on structural percentages rather than spatial lines, it easily handled our imbalanced data and caught the highest number of actual churning accounts (Highest Recall), proving that being on a Month-to-Month contract is the single biggest statistical trigger for leaving.
Support Vector Machine	**The Margin Maximizer.** This model tries to draw a wide, safe boundary separating loyal users from churning users. It performed beautifully because our feature engineering simplified the map for it: once we smoothed out the massive 'Total Charges' spikes using log math and grouped 'Tenure' into clear yearly milestones, the model easily separated the high-risk users from the loyal ones.

■ **Overall Production Winner Selection Justification: NAIVE BAYES** is chosen as the optimal operational framework, securing a peak validation F1-Score of **0.6157**.

Rationale: Customer churn telemetry is heavily class-imbalanced, causing geometric boundary models to get dragged toward the dominant class.Winning Model completely bypasses this issue by utilizing pure conditional percentage odds instead of geometric spatial lines. This structural focus allows it to isolate high-risk transactional milestones—such as Month-to-Month contracts—and maximize our capacity to catch actual at-risk accounts, delivering the highest overall model Recall profile.